

Yaoxuan Zeng

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EDUCATION

The University of Chicago, Department of the Geophysical Sciences Chicago, IL, USA

Ph.D. student in Prof. Malte F. Jansen's group Sep 2021 – June 2026 (expected)

- Thesis title: Ocean circulation on Enceladus and other icy moons

Peking University, School of Physics

Beijing, China

B.S. in Atmospheric and Oceanic Sciences

Sep 2016 – Jul 2020

- Working on “Oceanic superrotation on tidally locked planets” in Prof. Jun Yang's group

APPOINTMENTS

MIT School of Science Postdoc Fellow

Cambridge, MA, USA

Supervised by Prof. Wanying Kang

Sep 2026 (expected) –

Woods Hole Oceanographic Institution

Woods Hole, MA, USA

GFD fellow at the Geophysical Fluid Dynamics Program

Jun 2023 – Aug 2023

Peking University, School of Physics

Beijing, China

Research Assistant with Prof. Jun Yang

Sep 2020 – Jul 2021

PUBLICATIONS

1. **Yaoxuan Zeng**, and Malte F. Jansen: Interplay between slantwise convection and baroclinic eddies in the icy moon oceans. In preparation.
2. **Yaoxuan Zeng**, Wanying Kang, Glenn Flierl, and Geoff Vallis: Influence of jet depth on the equatorial jet direction on giant planets. Under review with *The Planetary Science Journal* (preprint: <https://arxiv.org/abs/2503.17828>).
3. **Yaoxuan Zeng**, and Malte F. Jansen: Slantwise convection and heat transport in icy moon oceans. *Geophysical Research Letters* 53, no. 3 (2026): e2025GL118761.
4. **Yaoxuan Zeng**, and Malte F. Jansen: Symmetric instability of Boussinesq fluid on a rotating planet. *Journal of Geophysical Research: Planets* 130, no. 7 (2025): e2025JE009115.
5. **Yaoxuan Zeng**, and Malte F. Jansen: The effect of salinity on ocean circulation and ice-ocean interaction on Enceladus, *The Planetary Science Journal* 5, no. 1 (2024): 13
6. Malte F. Jansen, Wanying Kang, Edwin Kite, and **Yaoxuan Zeng**: Energetic constraints on ocean circulation of icy ocean worlds, *The Planetary Science Journal* 4, no. 6 (2023): 117.
7. **Yaoxuan Zeng**, and Malte F. Jansen: Ocean Circulation on Enceladus with a High- versus Low-salinity Ocean, *The Planetary Science Journal*, 2021, 2, 4.
8. **Yaoxuan Zeng**, and Jun Yang: Oceanic Superrotation on Tidally Locked Planets, *The Astrophysical Journal*, 2021, 909, 172.
9. Jun Yang, Weiwen Ji and **Yaoxuan Zeng**: Transition from Eyeball to Snowball Driven by Sea-ice Drift on Tidally Locked Terrestrial Planets, *Nature Astronomy*, 2020, 4, 58-66.

SELECTED PRESENTATIONS

1. Slantwise convection and heat transport in icy moon oceans. 25th Conference on Atmospheric and Oceanic Fluid Dynamics, Houston, TX, USA, Jan 2026.
2. Slantwise convection in icy moon oceans. 2024 AGU Annual Meeting, Washinton DC, USA, Dec 2024.
3. Linking Potential Vorticity mixing and jets on giant planets (poster). 2024 AGU Annual Meeting, Washington DC, USA, Dec 2024.
4. Symmetric instability of Boussinesq fluid on a rotating planet. 24th Conference on Atmospheric and Oceanic Fluid Dynamics, Burlington, VT, USA, June 2024.
5. Ocean circulation on Enceladus: the effect of salinity on ocean heat and tracer transport. Purdue Storm Snack seminar, West Lafayette, IN, USA, Jan 2024.
6. The effect of ocean heat transport on Enceladus's ice geometry under different salinities. 2023 AGU Fall Meeting, San Francisco, CA, USA, Dec 2023.
7. The effect of ocean heat transport on Enceladus's ice geometry under different salinities. 55th DPS Meeting Joint With EPSC, San Antonio, TX, USA, Oct 2023.
8. Jets on Giant Planets – A Tale of Two Forcings. GFD fellow project presentation, WHOI, Woods Hole, MA, USA, Aug 2023.
9. The effect of gas bubbles on nanoparticle transport and ocean circulation on Enceladus. 2022 AGU Fall Meeting (poster), Chicago, IL, USA, Dec 2022.
10. Ocean Circulation on Enceladus with A High- Versus Low-salinity Ocean. 23rd Conference on Atmospheric and Oceanic Fluid Dynamics, Breckenridge, CO, USA, June 2022.
11. Wind-driven Ocean Circulation on Tidally Locked Aqua-planets. Astrobiology Science Conference, Bellevue, WA, USA, Jun 2019.

SELECTED AWARDS AND HONORS

- MIT School of Science Postdoctoral Fellowship 2026-2029
- **GFD fellowship** of WHOI 2023
- **First Place Outstanding Student Presentation Award: Oral** of American Meteorological Society, 23rd AOFD conference 2022
- **Neubauer Graduate Scholarship** of UChicago 2021-2026
- **Weiming** Scholar at PKU (awarded for excellent academic and research) 2020
- **Excellent graduate** of Beijing 2020

TEACHING AND SERVICE

- **Lecturer of Global Warming: Understanding and Forecast** 2021/22/24
- **Teaching Assistant of Geophysical Fluid Dynamics: Foundations** 2025
- **Teaching Assistant of Geophysical Fluid Dynamics: Rotation and Stratification** 2024/26
- **Core organizer** for [Rossbypalooza Summer School](#) at UChicago 2024
- **Organizer for GeoSci Grad Expo** at UChicago 2024
- **Co-organizer for Earth Science Day** for High School Students at UChicago 2023
- **Organizer for Climate Journal Club and Dynamics group meeting** 2022/23/24/25/26
- **Lecturer of Global Warming: Understanding and Forecast** 2021/2022/2024
- **Reviewer** for *Nature Communications: Earth & Environment*